Where to Get Filters for Recognition

There are Several, handcrafted filters in computer vision

For example:
Canny, Sobel, Gaussian blur, smoothing, low- level segmentation, morphological filters, Gabor filters,...

- **Are they optimal for recognition?**
- **Can we learn them from our data?**

Filter Parameters

$$\begin{bmatrix} \theta_1 & \theta_2 & \theta_3 \\ \theta_4 & \theta_5 & \theta_6 \\ \theta_7 & \theta_8 & \theta_9 \end{bmatrix}$$